Esti Blanco-Elorrieta

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Updated: January, 2026

ACADEMIC EMPLOYMENT

Assistant Professor: **New York University** (2023 -)
Department of Psychology
Department of Neural Science

Post-doctoral fellow: **Harvard University** (2020 – 2023)
Cognitive Neuropsychology Lab (PI: Alfonso Caramazza)
Department of Psychology

EDUCATION

Ph.D.: **New York University** (2015 – 2020)
Psychology (Supervisor: Liina Pylkkänen)
Thesis: *Towards an ecologically valid neuroscience of bilingualism*.

Harvard University (research fellow, 2018 – 2019)
Psychology (Supervisor: Alfonso Caramazza)

M.Sc.: **Basque Center on Cognition, Brain and Language** (Honors, 2013)
Cognitive Neuroscience of language.

B.A.: **University of Deusto** (Honors, 2012)
Basque philology and Linguistics

AWARDS AND SCHOLARSHIPS

2025: Early Career Award: Society for the Neurobiology of Language
2021: Glushko dissertation award: Cognitive Science Society.
2021: NYU-wide Outstanding dissertation award in the basic sciences.
2021: Graduate School of Arts and Sciences: Outstanding dissertation award.
2020: Ben Barres fellowship for post-doctoral research.
2020: Coons/Leibowitz graduate student teaching award.
2019: Forbes 30 under 30 in Science.
2019: Martin Braine Fellowship in Psychology.
2018: Dingwall Foundation Award in the Cognitive, Clinical, and Neural Foundations of Language.
2018: New York University: Dean's Student Travel Award.
2017: Society of the Neurobiology of Language: Predoctoral fellow merit award.
2017: Woodcock Institute Award for the advancement of neurocognitive research.
2016: Helmsley Fellowship for cross-disciplinary research.
2016: New York University: Dean's Student Travel Award.
2015: Neurobiology of Language Conference: Travel Award.
2015: New York University: MacCracken Fellowship for doctoral studies.
2014: La Caixa Foundation: fellowship for graduate studies in North America (4% success rate).

2012: BCBL: academic scholarship (awarded to the 5% most promising incoming students).
2011: Basque Government fellowship for undergraduate research projects.
2010: University of Deusto: Dean's fellowship for undergraduate research projects.
2008: Basque Government: Distinguished Academic scholarship (top 1% students graduating high school in the country).

GRANTS

Pending

2026 **From Quantum Sensors to Brain Science (Role: PI. Requested: $70,750)**
Funding Agency: NSF BCS

2026 – 2029 **Optimizing Meaning Across Languages: A Computational and Neural Investigation of Bilingual Semantic Structure (Role: PI. Requested: $706,750)**
Funding Agency: NSF BCS

Awarded

2025 – 2028 **Neural bases of bilingual language production (Role: PI $607,772)**
Funding Agency: NSF BCS (Grant 2446452)

2025 – 2028 **Graduate Research Fellowship (Role: Sponsor / Fellow: Maya Taliaferro. $144,500)**
Funding Agency: National Defense Science and Engineering Graduate Fellowship (NDSEG)

2024 – 2025 **Influence of multilingualism on brain plasticity and development (Role: PI $27,140)**
Funding Agency: NYU (Mega-grants initiative award)

2022 – 2027 **Language-specific and general mechanisms in bilingual aphasic individuals (Role: PI $1,003,338)**
Funding Agency: NIH/NIDCD (K99/R00, R00 DC019973-01)

2022 – 2025 **Adaptations to Multilingualism (Role: Advisory team 68.365,00 €).**
Funding Agency: Spanish Ministry of Science (PID2021-124399NA-I00).

JOURNAL PUBLICATIONS

Authorship note: * denotes trainee first author under my supervision

34. Taliaferro, M.*, Imel, N.*, Zavslasky, N.^, & **Blanco-Elorrieta, E.^ (***in prep*). Bilinguals exhibit semantic convergence while maintaining near-optimal efficiency.

33. Taliaferro, M.*, & **Blanco-Elorrieta, E.** (*under review*). Decoding the nature of conceptual representations in the bilingual mind.

32. Singh, A.*, Taliaferro, M.*, & **Blanco-Elorrieta, E.** (*under review*). Blending Boundaries: A Computational Approach to How Bilinguals Reconcile Cross-Linguistic Categorization.

31. Chen, X.*, Salvadore, M., & **Blanco-Elorrieta, E. (***under review*). A large dataset of bilingual data across four populations.

30. Arantzeta, M**.** & **Blanco-Elorrieta (***under review*). Absence of Treatment Generalization and Cross-Language Transfer in Bilingual Phonological Anomia. *Aphasiology*

29. Chen, X.*, Salvadore, M., & **Blanco-Elorrieta, E. (***under review*). A large sample investigation across 4 language groups reveals no consistent bilingualism specific neural adaptations.

28. Chen, X.*, & **Blanco-Elorrieta, E. (***under review*). A Shared Neural Mechanism for Morphological Inflection Across Languages.

27. Chen, X.*, Marte, M., Kiran, S., & **Blanco-Elorrieta, E. (***2026*). Evidence for an integrated bilingual language system across levels of representation through discourse tasks in bilingual aphasia. *Aphasiology,* 1-25.

26. Singh, A.*, Taliaferro, M.*, & **Blanco-Elorrieta, E.** (2025). Blending Boundaries: A Computational Approach to How Bilinguals Reconcile Cross-Linguistic Categorization. *Proceedings of the 46th Annual Meeting of the Cognitive Science Society.*

25. Taliaferro, M.*, Imel, N.*, Zaslavsky, N.^, & **Blanco-Elorrieta, E.^** (2025). Bilinguals exhibit semantic convergence while maintaining near-optimal efficiency. *Proceedings of the 46th Annual Meeting of the Cognitive Science Society.*

24. Chen, X.*, & **Blanco-Elorrieta, E. (***2025*). A theoretically driven and empirically grounded calculation for language dominance and degree of multilingualism. *Bilingualism: Language and Cognition,* 1-7.

23. **Blanco-Elorrieta, E.** & Arantzeta, M. **(***2025*). Basque-Spanish bilingual aphasia: a case-study to inform bilingual language organization. *Neurobiology of Language,* 1 - 35.

22. **Blanco-Elorrieta, E. (**2024). Is adaptation the new "bilingual advantage"?. *Trends in Cognitive Sciences.* 1-8.

21. Abdullah, L., **Blanco-Elorrieta, E,** & Valdez, D. L. (2024). "¿ Cómo qué, cómo qué? cómo qué?" Single-language echolalia in a bilingual female with progressive supranuclear palsy: a case report. *Neurocase*, 1-8.

20. Baus, C., Millan, I., Chen, X.*, & **Blanco-Elorrieta, E.** (2024). Exploring Language Comprehension through Cortical Tracking: the bilingual test case. *Neurobiology of Language,* 1-28.

19. Zhu, J., **Blanco-Elorrieta, E.,** Sun, Y., Szakay, A, & Sowman, P. (2022). Natural vs. forced language switching: free selection and consistent choices eliminate significant performance costs and cognitive demands in the brain. *NeuroImage,* 247, 188797.

18. **Blanco-Elorrieta, E**.*, Hauptman, M.*, & Pylkkänen, L. (2021). Inflection across categories: Tracking abstract morphological processing in language production with MEG. *Cerebral Cortex,* 309.

17. **Blanco-Elorrieta, E.** & Caramazza, A. (2021). On the need of theoretically guided approaches to possible bilingual advantages: an evaluation of the potential loci in the language and executive control systems. *Neurobiology of Language*, 1-16.

16. Honari-Jahromi, M., Chouinard, B., Fyshe, A., **Blanco-Elorrieta**, **E.,** & Pylkkänen, L. (2021). Neural representation of words within phrases:Temporal evolution of color-adjectives and object-nouns during simple composition. *Plos One*, 16(3), e0242754.

15. **Blanco-Elorrieta, E.** & Caramazza, A. (2021). A common selection mechanism at each linguistic level in bilingual and monolingual language production. *Cognition,* 104625.

14. **Blanco-Elorrieta, E*.,** Gwilliams, L*., Marantz, A., & Pylkkänen, L. (2020). Adaptation to mis-pronounced speech: evidence for a prefrontal-cortex repair mechanism. *Nature Scientific Reports,* 11, 97.

13. **Blanco-Elorrieta, E.,** Ding, N., Pylkkänen, L., & Poeppel, D. (2020). Understanding requires tracking: Noise and knowledge interact in bilingual language comprehension. *Journal of Cognitive Neuroscience*. 32(10), 1975-1983.

12. Liu, H., Zhang, Y., **Blanco-Elorrieta, E.,** Hea, Y., Chen, B. (2019). Role of proactive control on subcomponents of language control: Evidence from trilinguals. *Cognition, 194,* 104055.

11. **Blanco-Elorrieta, E.,** & Pylkkänen, L. (2018). Ecological validity in bilingualism research and the bilingual advantage. *Trends in Cognitive Sciences,* 22(12), 1117-1126

10. **Blanco-Elorrieta, E.,** Emmorey, K., & Pylkkänen, L. (2018). Language switching decomposed through MEG and evidence from bimodal bilinguals. *Proceedings of the National Academy of Sciences, 115(39),* 9708-9713.

9. **Blanco-Elorrieta, E.,** Kastner, I., Emmorey, K., & Pylkkänen, L. (2018). Shared neural correlates for building phrases in signed and spoken language. *Nature Scientific Reports,* 8, 5492.

8. **Blanco-Elorrieta, E.** Emmorey, K., & Pylkkänen, L. (2017). Decoding Language Switching in the Bilingual Brain: evidence from simultaneous speech and sign production. *Cognitive Computational Neuroscience (CCN)*.

7. Fyshe, A., **Blanco-Elorrieta, E**. & Pylkkänen, L. (2017). The neural representation of concepts during composition. Cognitive Computational Neuroscience (CCN).

6. **Blanco-Elorrieta, E.,** Ferreira, V.S., Del Prato, P., & Pylkkänen, L. (2017). The priming of basic combinatory responses in MEG. *Cognition,* 70, 49-63.

5. **Blanco-Elorrieta, E.** & Pylkkänen, L. (2017). Bilingual language switching in the lab vs. in the wild: The spatio-temporal dynamics of adaptive language control. *The Journal of Neuroscience*, 37: 9022-9036.

4. **Blanco-Elorrieta, E**. & Pylkkänen, L. (2016). Bilingual language control in perception vs. action: MEG reveals comprehension control mechanisms in anterior cingulate cortex and domain-general production control in dorsolateral prefrontal cortex. *The Journal of Neuroscience,* 36(2): 290 – 301.

3. **Blanco-Elorrieta, E.** & Pylkkänen, L. (2016). Composition of complex numbers: Delineating the computational role of the left anterior temporal lobe. *NeuroImage*, 124*, 194 - 203.*

2. **Blanco-Elorrieta, E.** & Pylkkänen, L. (2015). Brain bases of language selection: MEG evidence from Arabic-English bilingual language production. *Frontiers in Human Neuroscience*, 9.

1. Pylkkänen, L., Bemis, D.K. & **Blanco-Elorrieta, E**. (2014). Building phrases in language production: An MEG study of simple composition. *Cognition*, 133, 371-384.

INVITED TALKS [last 5 years]

Plenary

43. *Architectures and Mechanisms for Language Processing AMLaP Conference*, Donostia-San Sebastian, Basque Country. (August 2023).

42. *XVI International Symposium of Psycholinguistics*, Vitoria-Gasteiz, Basque Country. (May 2023; available here).

41. *Young Scholar Speaker Series*, Pennsylvania State University, USA. (March 2021).

Keynotes

40. *Neurolinguistics in Sweden Conference*. Stockholm, Sweden (June 2026).

39. *Bilingual Language Processing: Bridging Human and Artificial Cognition Conference*. France (April 2026).

38. *Early Career Award Recipient.* Society for the Neurobiology of Language, USA (September 2025).
37. *2nd Conference in Clinical Linguistics*. Baiona, France. (July 2025).
36. *LGBTQ+ Science Research Showcase*. UCL, London, UK. (June 2025).
35. *5th International Conference on Theoretical East Asian Psycholinguistics*. Tokyo, Japan. (April 2025).
34. *Adaptations to Multilingualism/Bilingualism and the Role of Linguistic Proximity.* Universidad Autonoma de Barcelona, Catalunya (February 2025).
33. *Stony Brook, Yale, NYU & CUNY (SYNC) Linguistics Conference*, USA. (December 2024).
32. *Perspectives on Multilingualism: An Interdisciplinary Roundtable.* NYUAD Research Institute, New York, USA. (January 2023; available here).
31. *Symposium on Dementia and Brain Aging in LMICs.* Nairobi, Kenya. (March 2022).
30. *Intersectional Experiences in Academia,* University of Oslo, Norway. (March 2022).
29. *NIH National Institute on Aging's Virtual Workshop*, USA. (March 2021).

Colloquiums

28. *Cognitive Science Colloquium*. Indiana University, USA (February 2026).
27. *Cognitive and Brain Science Lecture Series*. Tufts University, USA (November 2025).
26. *LOT Summer School on Linguistics*. University of Antwerp, Belgium. (June 2025).
25. *Psycholinguistics Workshop*. Tokyo, Waseda University, Japan. (April 2025).
24. *Concepts and Categories Colloquium*. NYU, USA (March 2025).
23. *Communicative Sciences and Disorders Colloquium*. NYU, USA (March 2025).
22. *Neuroscience Graduate Program Brown Bag.* UC San Diego, USA. (February 2025).
21. *Language and Communicative Disorders Speaker Series*. SDSU, USA. (February 2025).
20. *USA & Spain: Bridge for the Neurosciences Series*. King Juan Carlos Center, NY, USA. (October 2024).
19. *Speech-Language-Hearing Sciences Colloquium.* The Graduate Center of CUNY, USA. (October 2024).
18. *2025-2029 Strategic Plan Development Workshop*. University of Mondragon, Basque Country. (March 2024).
17. *Donostia International Physics Center Colloquium.* Donostia, Basque Country. (March 2023; available here).
16. *Cognitive Sciences Program Speaker Series.* Dartmouth University, USA. (October 2022).
15. *Psychology Department Seminar.* New York University, USA. (March 2022).
14. *Cognitive Science Seminar.* Johns Hopkins University, USA. (March 2022).
13. *Language and Cognitive Science Seminar.* Delaware University, USA. (March 2022).
12. *Princeton Neuroscience Institute Seminar.* Princeton University, USA. (March 2022).
11. *Neuroscience Research Seminar.* University of Texas Rio Grande Valley, USA. (February 2022).
10. *Cognitive Area Talk,* University of Michigan, Ann Arbor, USA. (January 2022).
9. *Aphasia Community Group Seminar Series*, Boston University, USA. (November 2021).
8. *Language and Brain Colloquium*, University of Florida, USA. (October 2021).
7. *Department of Language Science*, University of California Irvine, USA. (March 2021).
6. *Brain and Cognitive Sciences Department. MIT*, Cambridge, USA. (May 2020).

5. *Cognitive Science Department.* Dartmouth University, New Hampshire, USA. (May 2020).
4. *Linguistics Department Colloquium*. NYU, USA. (March 2020).
3. *Zuckerman Institute.* Columbia University, New York, USA. (March 2020).
2. *Entre cuatro paredes*. Expatriados & Co., New York, USA. (March 2020).
1. *Linguistics Department*. Cornell University, Ithaca, USA. (February 2020).

TEACHING

Teaching Instructor (NYU)

Spring 2026: Undergraduate Cognition
Spring 2024: Undergraduate Cognition
Fall 2024: Undergraduate Cognition

Guest Lectures & Summer Schools

2026: *Cold Spring Harbor Laboratories*. Week-long summer course on Genetics and Neurobiology of Language.
2025: *Graduate School of Linguistics (LOT)*. Week-long summer course on neurolinguistics (Radboud University).
2022: *Linguistics and Cognitive Science*. Lecture on the neural bases of bilingualism (NYU).
2020: *Cognitive Psychology*. Lecture on bilingualism and sign language cognition (Dartmouth).
2019: *Neural bases of Language:* Lecture on the neurobiology of bilingualism (NYU).
2017: *Neurolinguistics.* Lecture on bilingualism and aging (Young Leaders of the Americas Initiative).
2017: *Cognition.* Lecture on language and cognition (NYU).
2016: *Neural Bases of Language.* Lecture on the neurobiology of bilingualism (NYU).
2016: *Neural Bases of Language.* Lecture on the neural bases of perceptual attunement (NYU).

MEMBERSHIP IN PROFESSIONAL ORGANIZATIONS

2025 – 2028 Elected Treasurer, Society for the Neurobiology of Language
2020 – 2022 Accessibility and Diversity Committee, Society for the Neurobiology of Language
2018 – 2021 Elected Board Member, Society for the Neurobiology of Language
2016 – present Member, Society for Neuroscience
2015 – present Member, Cognitive Neuroscience Society
2014 – present Member, Society for the Neurobiology of Language

ADVISING AND MENTORING

PhD advisees

2024-present Aditi Singh (Psychology, NYU)
2023-present Jessica Chen, Maya Taliaferro (Psychology, NYU)

Undergraduate research projects

2025 Anushree Anandani (Psychology, NYU)
2022-2023 Alisa Ragassa (Psychology, Harvard)

Dissertation committee

2026 Alline-Priscillia Messi (Psychology, NYU),
Jessica Goebbel (Linguistics, NYU)
2024 Natália Pinheiro (Linguistics, Federal University of Santa Catarina in Brazil)

NYU qualifying paper committee

Jessica Goebbel (Linguistics, NYU)
Auromita Mitra (Linguistics, NYU)

First and Second year paper committee

Hanxiao Lu (Psychology, NYU)
Meghan Kalvey (Psychology, NYU)
Youjing Luo (Psychology, NYUAD)

SERVICE

a) Departmental service

Center for Brain Imaging Protocol Review Committee
Diversity Equity and Inclusion Committee (now Community Building and Outreach Committee)
Education Policy Committee (EPC)

b) Ad hoc journal reviewer

Nature Reviews Psychology; Neurobiology of Language; Cognition; Scientific Reports; Human Brain Mapping; Proceedings of the National Academy of Sciences, PLOS Biology; PLOS One; Journal of Neuroscience; NeuroImage; Brain and Language; Language, Cognition and Neuroscience; Journal of Cognitive Psychology; Journal of Neurolinguistics, Bilingualism: Language and Cognition; Cerebral Cortex; Journal of Cognitive Neuroscience.

c) Ad hoc grant reviewer

National Science Foundation (NSF) – Linguistics and Developmental Sciences Program Panel; Hong Kong Research Grants Council; Polish Ministry of Science and Higher Education; Polish National Centre for Research and Development; Rochester Neuroscience Grant Program.

d**) Ad hoc conference reviewer**

Architectures and Mechanisms for Language Processing (AMLaP), Society for the Neurobiology of Language (SNL), International Symposium on Bilingualism (ISP), International Conference on Theoretical East Asian Psycholinguistics**.**

e) Conference organization

Program committee (Society for the Neurobiology of Language annual meeting; 2020, 2021, 2026).

f) Elected Representative

Society for the Neurobiology of Language: Treasurer (2026-2028).
Society for the Neurobiology of Language: Student and Post-doc representative (2019-2022).

SCIENTIFIC OUTREACH

2026: Interview at Academic Minute, WAMC Northeast Public Radio, NPR, USA.
2025: Interview at ABC News, USA.
2024: Interview at "The Science of Perception Box" podcast, USA.
2024: Outreach Talk at "Sin Fronteras" Educational in Madrid, Spain.
2023: Outreach Talk on Bilingualism and Child Development at Montessori School in Williamsburg, NYC.
2023: Interview in La Mecanica del Caracol radio program on language policy (here, in Spanish).
2023: Interview in El Diario Vasco newspaper on language evolution (here, in Spanish).
2023: Interview on Berria newspaper about the prejudices around multilingualism (here, in Basque).
2023: Podcast episode on the psychology, neuroscience and philosophy of multilingualism (here, in English).
2023: 'De ondas de sonido a significado' Outreach talk at the University of the Basque Country.
2023: Radio program at EITB discussing the bilingual brain (available here in Spanish).
2023: 'How Does the Bilingual Brain Process Language?' Public talk at NYUAD Institute (here, in English).
2023: 'Como funciona el cerebro bilingüe?' Talk at San Telmo Museum *NanoNeuro* series (here, in Spanish).
2022: TV Interview on the effects on multilingualism on the brain in EITB (available here, in Basque).
2022: Interview on multilingualism and language policies in El Español newspaper (available here, in Spanish).
2022: Interview at 'Gure artean' podcast, EiTB.
2021: Panelist and Organizer, Early Career Event, Society for the Neurobiology of Language.
2021: Speaker, Doctoral Student Welcome Event, New York University.
2021: TED en Español Podcast episode (available here, in Spanish).
2021: Interview in Archiletras bi-monthly scientific magazine (available here, in Spanish).
2021: Baipasa: weekly contribution to science radio program (available here, in Basque).
2020: Interview in a TV episode of 'The brain hunter' (available here, in Spanish).
2020: TED Talk in Spain, 2020 (available here, in Spanish).
2020: Interview on the Neurobiology of multilingualism on COPE radio (available here, in Spanish).
2020: Interview on the architecture of the multilingual brain in RTVE Spanish radio (available here, in Spanish).
2019: '*How to build a career in Science'* panelist; Young Leaders of the America Initiative (YLAI).
2019: '*How does switching between languages impact your body?*' Inspire series by Euronews (here in English).

(TRANS)GENDER ADVOCACY

2025: Keynote Speaker at University College London: LGBTQ+ Research Showcase.
2024: Interview on TODAY newspaper on gender affirming voice therapy (available here, in English).
2022: University of Oslo: Intersectional Experiences in Academia workshop. Panelist and workshop leader.
2021: Spanish ministry for Gender Equality: advisory committee.
2021: *'Identity, language, and the future'* Twitch interview for 'The Future Game' (available here, in Basque).
2021: Bayer pharmaceutical Latin-American branch: Pride event panelist.
2021: '*Non-binary'* feature documentary on trans experience (available here, English subtitles).
2020: ELA Basque Trade Union: Pride event panelist.
2020: National Organization of Gay and Lesbian Scientists and Technical Professionals: Speaker, career panel.
2020: Harvard Women in STEM Initiative, Mentor.
2019: The Lesbian, Gay, Bisexual & Transgender Community Center: Panelist, 'Being trans* in Academia.'

MEDIA COVERAGE (selected)

2025: Coverage of our bilingualism measure in IFL Science (here, in English).

2023: Newspaper interview in *Diario Vasco* on language and evolution (here, in Spanish).
2023: Radio interview at *Radio Vitoria* on how the human brain processes language (here, in Spanish).
2023: Interview on *Nius Diario* on the myths around bilingualism and the brain (here, in Spanish).
2023: Interview in *The New York Times* on bilingualism and aging (here, in English).
2023: Interview in *Berria* newspaper on the bilingual brain (here, in Basque).
2023: Interview in *ARGIA* magazine on the bilingual brain (here, in Basque).
2020: 'Bilingual individuals' interview in El Pais newspaper (available here, in Spanish)
2020: *'Genius of a thousand languages'* interview in Deia newspaper (available here, in Spanish)
2020: Interview on Multilingualism and being a Forbes Scientist, on EITB radio (available here, in Basque)
2020: *'How does one become a Forbes Scientist?'* interview on ONDA CERO radio; here in Spanish)
2019: Coverage and Interview in the *Inspire* series by Euronews (available here in English)
2018: *'When we sign, we build phrases with similar neural mechanisms as when we speak'* (American Association for the Advancement of Science, available here in English)
2018: *'Uno stesso circuito per lingua parlata e lingua dei segni'*, Le Scienze magazine, available here, in Italian)
2018: *'Four amazing discoveries you probably didn't hear about this week'*, NSF podcast, here, in English)
2018: *'How we turn on and off languages'*, Science Daily (available here, in English)
2017: *'Chatting makes it easy to switch languages'*, Futurity news (available here, in English)

LANGUAGES

Basque/Spanish: Native speaker. **English:** Near native. **German:** Conversational.